(18) How many st lines can be formed with 13 non-collinear points?

Ans:- to form a st line we must need two non-collineas points. like as shown =>

So, that 2 points of non-collinear to be selected means combinations.

$\Rightarrow$ Total st lines $= {}^{13}C_2 = \frac{13 \times 12 \times 11!}{11! \cdot 2!} = 78$ st lines.

(19) How many st lines can be formed with 20 points on them 7 are collinear & remaining all non-collinear?

Ans:- With non-collinear points only we can form st lines but we should form st lines with all points first & from them whatever st lines with collinear points we have taken should substracted from these all st lines.

Total st lines $= {}^{20}C_2$

st lines to be substracted with collinear $= {}^{7}C_2$

Required st lines $= {}^{20}C_2 - {}^{7}C_2 = \frac{20 \times 19}{2} - \frac{7 \times 6}{2}$

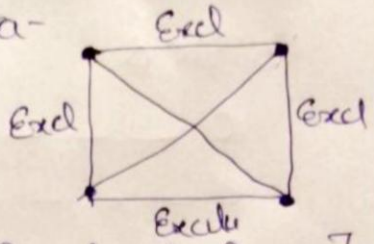
$= 190 - 21 = 169$ st lines.

(20) The total no. of diagonals in a 4 sided closed figure i.e., either square / rectangle / Quadrilateral / Parallelogram / Rhombus / Trapezium?

(234)

Ans:-

↳ Generally everyone think that to form diagonal we need two vertex & we have to select two vertex & how many joinings we may get that many diagonals is the answer, but in this total count sides are also counted. So, we have to substract them & take the result. Means, substra-ct the sides from total joinings.



$n C_2 - n$ [n is no. of sides in the closed figure]

$= \frac{n(n-1)}{2} - n = \frac{n(n-1)-2n}{2} = \frac{n(n-1-2)}{2} = \frac{n(n-3)}{2}$

No. of diagonals in 'n' sided figure = $\frac{n(n-3)}{2}$

here square/... $n=4$ =>

$\therefore$ diagonals $= {}^4C_2 - 4 = \frac{4\times3}{2} - 4 = 6-4 = \boxed{2}$

(or) no. of diagonals = $\frac{n(n-3)}{2} = \frac{4(4-3)}{2} = \boxed{2}$

(21) In how many ways 10 beads can be arranged to a necklace?

Ans:- here arrangement is circular & clock wise & anti-clock wise are considered as same.

$\therefore$ Total arrangements $= \frac{1}{2}(n-1)! = \frac{1}{2}\times 9!$

$= \frac{1}{2}\times 362880 = 1,81,440$ arrangements.

(22) What is rank of the word "vector" according to dictionary order?

Ans:- first arrange all the letters in alphabetical order

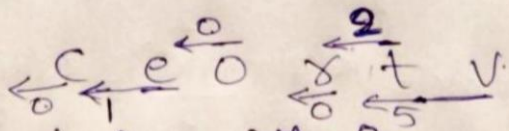
c e o r t v

(i) vector starting letter is 'v'. So, before v as a starting letter how many words completed we have to find.

c _ _ _ _ _ → 5!
e _ _ _ _ _ → 5!
o _ _ _ _ _ → 5!
r _ _ _ _ _ → 5!
t _ _ _ _ _ → 5!
v c _ _ _ _ → 4!
v e c o _ _ → 2!
v e c r _ _ → 2!
v e c t o r → 1.

∴ Rank is
$5(5!)+1(4!)+2(2!)+1$
$= 5(120)+24+2(2)+1$
$= 600+24+4+1$
$= 629$

Short cut:-



Our word starting letter is v before of how many letters & how many places vacant tables filling that letter we have to consider.

∴ 5 (5!)
↓ ↘
before letters 5 letters remaining in 5 places in 5! ways

$5(5!)+1(4!)+2(2!)+1$
$= 629$ th rank is vector.